Fig. 4.36(b) adds a reference gain H_r and a feedforward path $H_f s$ around the original unity-feedback loop. Zero steady-state error to a ramp requires both equalities

$$H_r = 1, \qquad H_f = \frac{1}{A} = 0.1. \tag{1}$$

For the unit parabola $r(t) = t^2/2$, the steady-state error is then τ/A, so the acceleration constant is

$$K_a = \frac{A}{\tau} = 10. \tag{2}$$

Part (c) is negative. Any departure of H_f from $1/A$ leaves a nonzero constant ramp error, and the system falls back to Type 1. H_r is stricter: any departure from 1 leaves no finite steady-state error, and the error grows at a constant slope. The simulation below, with $A = 10$, $\tau = 1$, and $r_o = 1$, sizes both effects and turns them into tolerance bands that can be written into a specification. The script is `type2_feedforward_robustness.m`; the printed table is `steady_state_results.txt`.

1 Error transfer function and the ideal Type 2

The plant is

$$G(s) = \frac{A}{s(\tau s + 1)}.$$

Both H_r and $H_f s$ enter on the reference path, and the feedback is unity negative feedback, so the loop gain is still $G(s)$. The characteristic polynomial

$$s(\tau s + 1) + A = \tau s^2 + s + A \tag{3}$$

does not contain H_r or H_f. Those two parameters change only the numerator of the error, not the poles.

The tracking error $E = R - Y$ has the transfer function

$$\frac{E(s)}{R(s)} = \frac{\tau s^2 + (1 - AH_f)s + A(1 - H_r)}{s(\tau s + 1) + A}. \tag{4}$$

The ramp reference is $r(t) = r_o t$, so $R(s) = r_o/s^2$. The numerator at $s = 0$ is $A(1 - H_r)$, and the coefficient of s is $1 - AH_f$.

The type for reference inputs is the number of zeros of this numerator at the origin, not the closed-loop poles:

- If $H_r \neq 1$, the numerator has no zero at the origin. The step error is $1 - H_r$, and the ramp error grows linearly in time. For references this is Type 0.
- If $H_r = 1$ but $H_f \neq 1/A$, the numerator has one zero at $s = 0$. The step error is 0 and the ramp error settles to a constant. This is Type 1.
- If both equalities hold, the numerator is only τs^2, a double zero at the origin. The ramp steady-state error is 0. This is the Type 2 the problem asks for.

The last two steady-state values follow from the final-value theorem. With $H_r = 1$ the numerator contributes one factor of s,

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{s\big[\tau s + (1 - AH_f)\big]}{\tau s^2 + s + A} \cdot \frac{r_o}{s^2} = r_o\Big(\frac{1}{A} - H_f\Big). \tag{5}$$

The ideal point $H_f = 1/A$ makes this zero.

If $H_r \neq 1$, the dc gain of E/R is $1 - H_r$, so a ramp remains a ramp. After the transient dies,

$$e(t) \to (1 - H_r)\, r_o\, t + r_o\Big(\frac{H_r}{A} - H_f\Big). \tag{6}$$

The first term is the slope. The second term is the constant that lines the formula up with the simulated curve. When $H_r = 1$ the slope vanishes and (6) reduces to (5).

At the ideal parameters, $E/R = \tau s^2/(\tau s^2 + s + A)$. The unit parabola $r(t) = t^2/2$ transforms to $1/s^3$, and its steady-state error is

$$e_{ss} = \lim_{s\to 0} \frac{\tau s^2}{\tau s^2 + s + A} \cdot \frac{1}{s^2} = \frac{\tau}{A}.$$

Defining $e_{ss} = 1/K_a$ gives (2). This K_a is defined from the reference error. The loop itself still has only the one integrator in $G(s)$, and $\lim_{s\to 0} s^2 G(s) = 0$. The extra order is a cancellation of numerator coefficients, not a second integrator in the loop.

The nominal poles come from $s^2 + s + 10 = 0$: $\omega_n = \sqrt{10} \approx 3.16$ rad/s and $\zeta = 1/(2\sqrt{10}) \approx 0.158$. The oscillation period is about 2 s, and the envelope decays as $e^{-t/2}$, so the transient is gone after about 8 to 10 s. Offsets in H_r and H_f alone do not move these poles.

2 Simulation results

The simulation fixes $A = 10$, $\tau = 1$, and $r_o = 1$. The ideal controller is (1). The ramp is applied to (4) with `lsim` from 0 to 30 s at a step of 0.01 s. Each offset is a fraction of the ideal value, $H = H_{\text{ideal}}(1 + \delta)$, with δ equal to 0, +1%, +5%, and +10%. The two parameters are moved separately: H_f is held at 0.1 while H_r moves, then H_r is held at 1 while H_f moves.

The value at 30 s has to be read in two ways. When $H_r = 1$ the transient is gone, so $e(30)$ is e_{ss}. When $H_r \neq 1$ the error is still climbing at a constant slope, so $e(30)$ is only the value at that instant and keeps changing if the run is longer. The column "asymptotic $e(30)$" is (6) evaluated at $t = 30$. Over the eight cases, the simulated endpoint and this asymptote differ by at most 6.071×10^{-8}. That residual is the same order as $e^{-30/2}$: leftover transient, not an error in the formula. The curves are Fig. 1. The endpoints match `steady_state_results.txt`.

2.1 Offset in H_r

H_f stays at 0.1. The numbers are Table 1; the curves are the upper plot of Fig. 1.

Table 1: $H_f = 0.1$ fixed, H_r above the ideal value. Slope and asymptote from (6), with $r(30) = 30$.

Offset	H_r	Slope	Asymptotic $e(30)$	Simulated $e(30)$	$e(30)/r(30)$
0%	1.00	0	0	0	0
+1%	1.01	−0.010	−0.299	−0.299	−1.00%
+5%	1.05	−0.050	−1.495	−1.495	−4.98%
+10%	1.10	−0.100	−2.990	−2.990	−9.97%

The ideal curve sits on 0 after about 10 s. The other three become straight lines after the same transient, with slope $-(H_r - 1)$. On the $+10\%$ curve the error goes from about -1 near 10 s to -2.99 at 30 s, a drop of about 2 in 20 s, which matches the slope -0.1.

$e(30)$ is slightly less negative than "slope $\times 30$". The difference is the constant term in (6). For $+1\%$, the slope term is -0.300 and the constant $r_o(H_r/A - H_f) = 0.001$, which add to -0.299. The larger the offset, the less this constant matters next to the slope term. For large t, the relative error therefore satisfies

$$\frac{e(t)}{r(t)} \to 1 - H_r.$$

If H_r is high by a few percent, the ramp is led by about that same percentage for the rest of the run. $e < 0$ means $y > r$: the reference gain is high, the command into the plant is strong, and the output runs ahead of the reference.

2.2 Offset in H_f

H_r stays at 1. The numbers are Table 2; the curves are the lower plot of Fig. 1.

Table 2: $H_r = 1$ fixed, H_f above the ideal value. e_{ss} from (5).

Offset	H_f	AH_f	Theoretical e_{ss}	Simulated $e(30)$
0%	0.100	1.00	0	0
+1%	0.101	1.01	−0.001	−0.001
+5%	0.105	1.05	−0.005	−0.005
+10%	0.110	1.10	−0.010	−0.010

The four transients lie almost on top of each other, with a peak near $+0.25$ and a valley near -0.16, because the poles did not move. The steady-state separation is only 0.001 to 0.010, and on a common vertical scale the transient hides it. The zoom from $t = 20$ to 30 s shows four horizontal lines at the e_{ss} values of Table 2.

From (5), if $H_f = (1/A)(1 + \beta)$,

$$e_{ss} = -\frac{r_o}{A}\,\beta = -0.1\,\beta. \tag{7}$$

$\beta = +0.10$ gives $e_{ss} = -0.010$, the same as the simulation. The error is a constant, so the relative error $e(t)/r(t)$ goes to 0 as $1/t$. At 30 s the $+10\%$ relative error is only $-0.010/30 \approx -0.033\%$. Next to the percent-for-percent H_r error of the previous subsection, the H_f penalty is smaller by a factor of order At.

The sign has the same origin: a high H_f injects too much velocity command through the derivative feedforward, and the output leads the reference by a fixed distance. The formula is linear in a

negative offset. Lowering H_f by 10% ($H_f = 0.09$) would give $e_{ss} = +0.010$, a lag. This run only swept the high side.

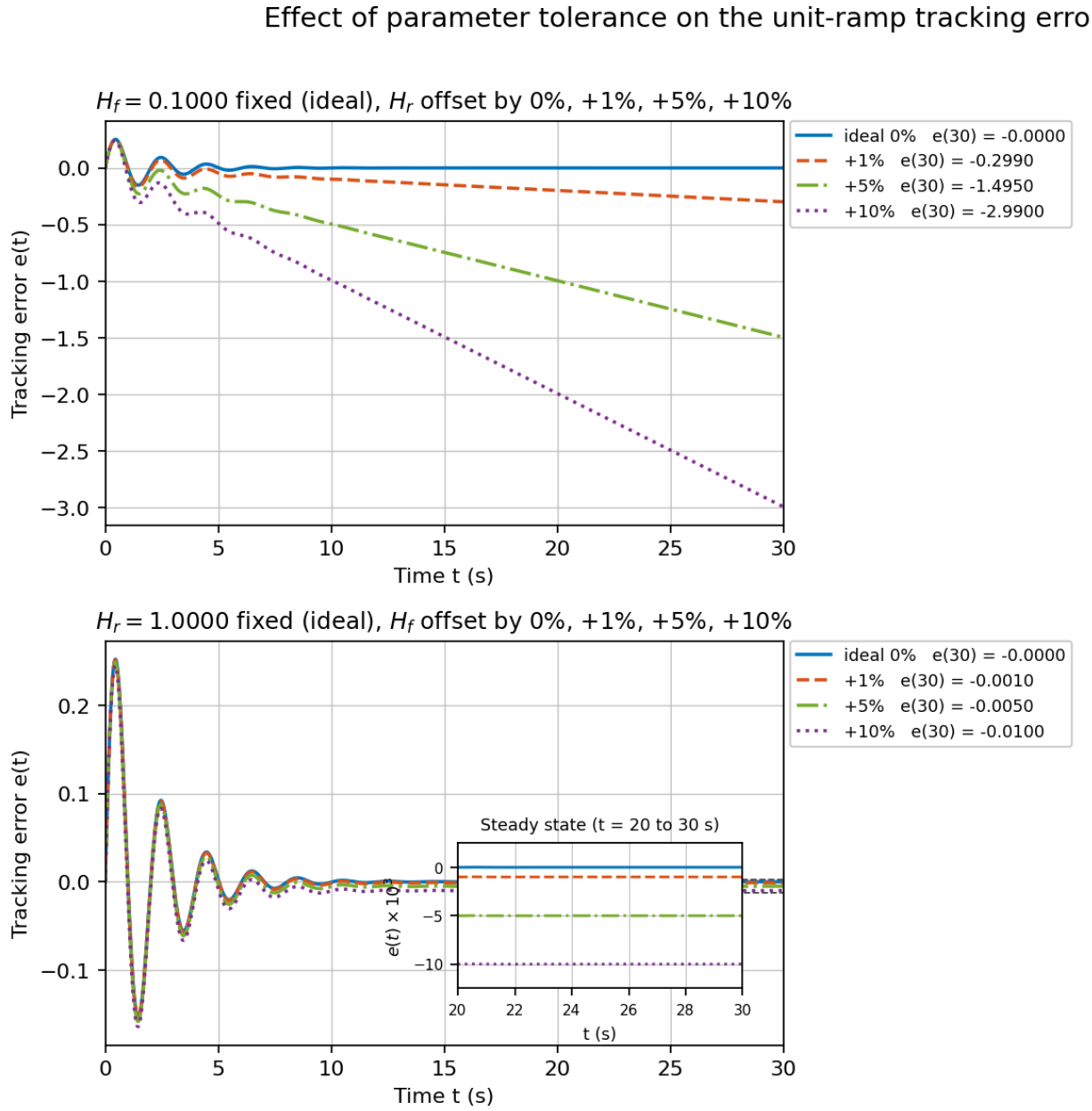


Figure 1: Tracking error $e(t) = r(t) - y(t)$ for the unit ramp $r(t) = t$. The upper plot holds $H_f = 0.1$; the lower plot holds $H_r = 1$. The inset is $t = 20$ to 30 s, with the vertical axis in units of 10^{-3}. The legend value $e(30)$ is the simulated endpoint.

3 Robustness and allowable error

3.1 Type 2 has no tolerance band

Part (c) asks whether the property survives, not how large the error is. Type 2 requires the ramp steady-state error to be exactly 0, which is the pair of equalities in (1). Any departure of H_f makes (5) nonzero, and the system returns to Type 1. The $+1\%$ row of Table 2 already gives $e_{ss} = -0.001 \neq 0$. This Type 2 is not robust to H_f, and there is no interval on which a small change still

counts as Type 2.

The structural reason is that the extra order comes from the cancellation $1 - AH_f = 0$. The loop gain does not contain a second integrator. If either side of the cancellation drifts, the zero leaves the origin. The closed-loop roots do not show it. The four H_f curves oscillate the same way and have the same stability margins, but the ramp error is no longer 0. Checking only the poles, or only a short transient, misses this tolerance.

The equality for H_r has no width either. Once it moves, even the Type 1 property that the ramp error stays bounded is lost, and the system is Type 0 for references. When feedforward is used to create the type, both the reference gain and the feedforward gain are hard equalities.

3.2 When the specification is an error bound

A product specification usually does not say "must be Type 2". It says the error must not exceed a number. A tolerance band can then be computed, and the rules for H_r and H_f are different. Set

$$H_r = 1 + \alpha, \qquad H_f = \frac{1}{A}(1 + \beta),$$

where α and β are relative errors. Substitution into (6) gives

$$e(t) \to -\alpha\, r_o\, t + \frac{r_o}{A}(\alpha - \beta). \tag{8}$$

H_r. If $\alpha \neq 0$, the error is unbounded. No nonzero tolerance on H_r keeps the ramp error finite for all time. If the requirement is only $|e(t)| \leq e_{\max}$ over a mission of length T, the slope term dominates,

$$|\alpha| \leq \frac{e_{\max}}{r_o T}. \tag{9}$$

A longer T demands a more accurate H_r. For the numbers here, $r_o = 1$, $e_{\max} = 0.30$, and $T = 30$ s, one gets $|\alpha| \leq 0.01$, which is the $+1\%$ row of Table 1 (the actual $|e(30)|$ is 0.299). The same budget of 0.30 carried out to 300 s forces $|\alpha|$ down to 0.001, that is 0.1%.

If the specification is a relative error $|e/r| \leq \varepsilon$, waiting longer does not help. From (8),

$$|\alpha| \leq \varepsilon. \tag{10}$$

The relative tolerance on H_r equals the allowed relative tracking error. A 1% ramp-tracking spec requires H_r within 1%, over the whole mission, not at one instant.

H_f. With $\alpha = 0$ the error is bounded,

$$|e_{ss}| = \frac{r_o}{A}\,|\beta| \quad \Rightarrow \quad |\beta| \leq \frac{A}{r_o}\, e_{\max}. \tag{11}$$

For $A = 10$ and $r_o = 1$:

Allowed $\lvert e_{ss}\rvert$	$\lvert H_f - 0.1\rvert$	Relative tolerance $\lvert\beta\rvert$
0.001	0.001	1%
0.005	0.005	5%
0.010	0.010	10%

The last row is the $+10\%$ case of Table 2. The relative error shrinks as $1/t$, so a few percent on H_f is usually acceptable when only long-term relative accuracy matters. When the absolute lag matters, use (11) instead of copying the H_r percentage.

Plant-gain drift. The simulation holds A fixed. In practice the condition is the product $AH_f = 1$, so drift in A and drift in H_f enter the same expression. With $H_r = 1$ and H_f left at its nominal value $1/A$, an actual gain $A(1+\gamma)$ gives

$$e_{ss} = r_o\Big(\frac{1}{A(1+\gamma)} - \frac{1}{A}\Big) = -\frac{r_o}{A}\cdot\frac{\gamma}{1+\gamma} \approx -\frac{r_o}{A}\,\gamma.$$

To first order, a 10% error in A and a 10% error in H_f are about the same size, and $|e_{ss}|$ is about 0.01. Setting the feedforward gain accurately is not enough: identification error in the plant gain comes in through the same path. This follows from (5) and was not simulated separately.

3.3 How to write the specification

If the specification is literally zero ramp error, neither H_r nor AH_f has a tolerance. This Type 2 is not a property that can be manufactured. Zero error requires the integrators to sit inside the loop, so that the type follows the loop gain instead of a feedforward cancellation.

If the specification is an error bound, the two parameters are written separately:

1. For H_r, use (10) or (9). A relative tracking error of a few percent means the same few percent on H_r. A longer mission tightens the absolute-error budget further. Any nonzero deviation has no steady state.

2. For H_f, and for the plant gain A, use (11). Divide the allowed constant lag by r_o/A to get the relative tolerance. Here $r_o/A = 0.1$, so a constant-error budget of 0.01 allows $\pm 10\%$ on H_f.

3. Acceptance has to look at the ramp error after the transient, not at a step response or at the poles. A steady H_f error of order 0.01 is hidden by a transient peak of about 0.25. The record has to cover an interval after the transient has died, with the tail enlarged as in the lower plot of Fig. 1.

On these numbers, 1% on H_r produces an error of -0.299 at 30 s, while 1% on H_f produces only -0.001. The same 1% component tolerance is about $At = 300$ times tighter for the reference gain than for the feedforward gain. The feedforward path can absorb a few percent of manufacturing spread. The reference gain cannot.